

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		Boxes 2 and 3 ticked: (1) + (1) Note – for each extra tick subtract 1 mark. No negative mark.		2		2		
		(ii)		Kepler 11 has <u>same</u> mass as our Sun (1) Kepler 11 has the [most] <u>similar</u> temperature to our Sun (1) Each additional item of data stated – 1 (minimum 0)			2	2		
	(b)	(i)		The exoplanet blocks / absorbs some light from the star Accept: there is a shadow / eclipse 'The planet doesn't give out light' – not enough	1			1		
		(ii)	I	2 orbits take 150 [days] or 300-150 [days] (1) Single orbit = $\frac{150}{2} = 75$ [days] (1) [Accept 75 – 78 days] Alternative 4 orbits take 300 [days] (1) [accept 310 days] Single orbit = $\frac{300}{4} = 75$ [days] (1) [Accept 75 – 78 days]			2	2	2	
			II	An extra/different / <u>smaller</u> dip in the intensity line is present [accept: reference to an anomaly] or there is a dip at ~160 days Not just: there is a slight dip Not: another transit shown			1	1		
			III	[An absorption spectrum arises because] gases absorb some light (1) Different gases [accept: elements] [in the planet's atmosphere] have <u>different</u> black lines / [absorb light at] different wavelength / frequencies. (1)	2			2		

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	(c)	(i)		In <u>both</u> systems the temperature decreases with distance / orbit radius (1) But In the Solar System Venus is the 2nd planet but the hottest /Mercury is the first and not the hottest (1) [i.e the nature of the anomaly must be made clear]			2	2		
		(ii)		0.25 – 0.26		1		1	1	
				Question 2 total	3	3	7	13	3	0